

Hardware Requirements and Justification for Continuous Cuffless Blood Pressure Monitoring System

1. Project Overview

The objective of this project is to develop a continuous cuffless blood pressure monitoring system using Pulse Transit Time (PTT). The system utilizes synchronized ECG and PPG signals acquired from the MAX86150 biosensor. The acquired physiological signals are processed using the ESP32 microcontroller to estimate blood pressure in real time.

2. Hardware Components Required

2.1 MAX86150 Integrated ECG and PPG Biosensor

Function

The MAX86150 is an integrated biomedical sensor capable of simultaneously acquiring Electrocardiogram (ECG) and Photoplethysmography (PPG) signals.

Why it is Required

- Provides synchronized ECG and PPG signals.
- Eliminates the need for separate ECG and PPG sensors.
- Enables Pulse Transit Time (PTT) calculation.
- Compact and suitable for wearable healthcare applications.
- Low power consumption for continuous monitoring.

Role in the System

- ECG signal is used to detect R-peaks.
 - PPG signal is used to detect pulse arrival.
 - PTT is calculated using the time difference between ECG and PPG signals.
 - Blood pressure estimation is performed using PTT values.
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2.2 ESP32 Microcontroller

Function

The ESP32 serves as the central processing and communication unit.

Why it is Required

- Provides sufficient processing capability for real-time biomedical signal analysis.
- Supports wireless communication through Wi-Fi and Bluetooth.
- Offers low power consumption and multiple communication interfaces.

Role in the System

- Reads ECG and PPG data from MAX86150.
 - Performs filtering and signal conditioning.
 - Detects ECG and PPG peaks.
 - Calculates Pulse Transit Time (PTT).
 - Executes blood pressure estimation algorithms.
 - Communicates with display devices and future mobile applications.
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2.4 Lithium-Ion Battery (3.7V)

Function

Portable power source for the wearable system.

Why it is Required

- Supports continuous operation without external power.
 - Compact and lightweight.
 - Suitable for wearable healthcare devices.
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2.5 TP4056 Battery Charging and Protection Module

Function

Battery charging and protection circuit.

Why it is Required

- Provides safe charging of the lithium-ion battery.
 - Protects against overcharging and over-discharging.
 - Improves overall system reliability and safety.
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2.6 Voltage Regulation Circuit

Function

Provides stable operating voltage to all electronic modules.

Why it is Required

- Battery voltage varies during charging and discharging.
- Sensors and microcontrollers require a stable supply voltage.
- Prevents measurement errors caused by power fluctuations.

Possible Implementation

- Buck Converter
 - 3.3V Voltage Regulator
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2.7 Breadboard (if prototype)/ PCB

Function

Circuit assembly and hardware integration.

Why it is Required

- Enables prototyping and testing.
 - Supports future migration to a compact PCB design.
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2.8 Jumper Wires and Connectors

Function

Electrical interconnection between modules.

Why it is Required

- Establishes communication and power connections.
 - Facilitates rapid prototyping and debugging.
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3.Motion Artifact Handling

Motion artifacts occur when the user moves their hand, changes posture, or walks while the measurement is being taken. These movements can introduce noise into the ECG and PPG signals and affect blood pressure estimation.

To reduce this problem, the system continuously checks the quality of the ECG and PPG signals obtained from the MAX86150 sensor. If abnormal fluctuations, excessive noise, or distorted waveforms are detected, the current measurement is ignored and blood pressure estimation is temporarily paused. Once a stable signal is detected, the system automatically resumes blood pressure calculation.

This approach improves the accuracy and reliability of the proposed blood pressure monitoring system.

SUMMARY:

S.No	Component	Purpose
1	MAX86150	ECG and PPG Signal Acquisition
2	ESP32	Signal Processing and Wireless Communication
3	3.7V Li-ion Battery	Portable Power Source
4	TP4056 Charging Module	Battery Charging and Protection
5	Voltage Regulator / Buck Converter	Stable Power Supply
6	Breadboard / PCB	Hardware Integration
7	Jumper Wires and Connectors	Electrical Connections

4.Conclusion

The proposed hardware architecture provides a compact, low-power, and scalable solution for continuous cuffless blood pressure monitoring. The MAX86150 enables synchronized ECG and PPG acquisition for Pulse Transit Time calculation, while the ESP32 performs real-time signal processing and blood pressure estimation. Motion artifacts are addressed through signal quality assessment techniques in the current implementation, with provisions for future enhancement using motion sensor-based adaptive filtering methods.

